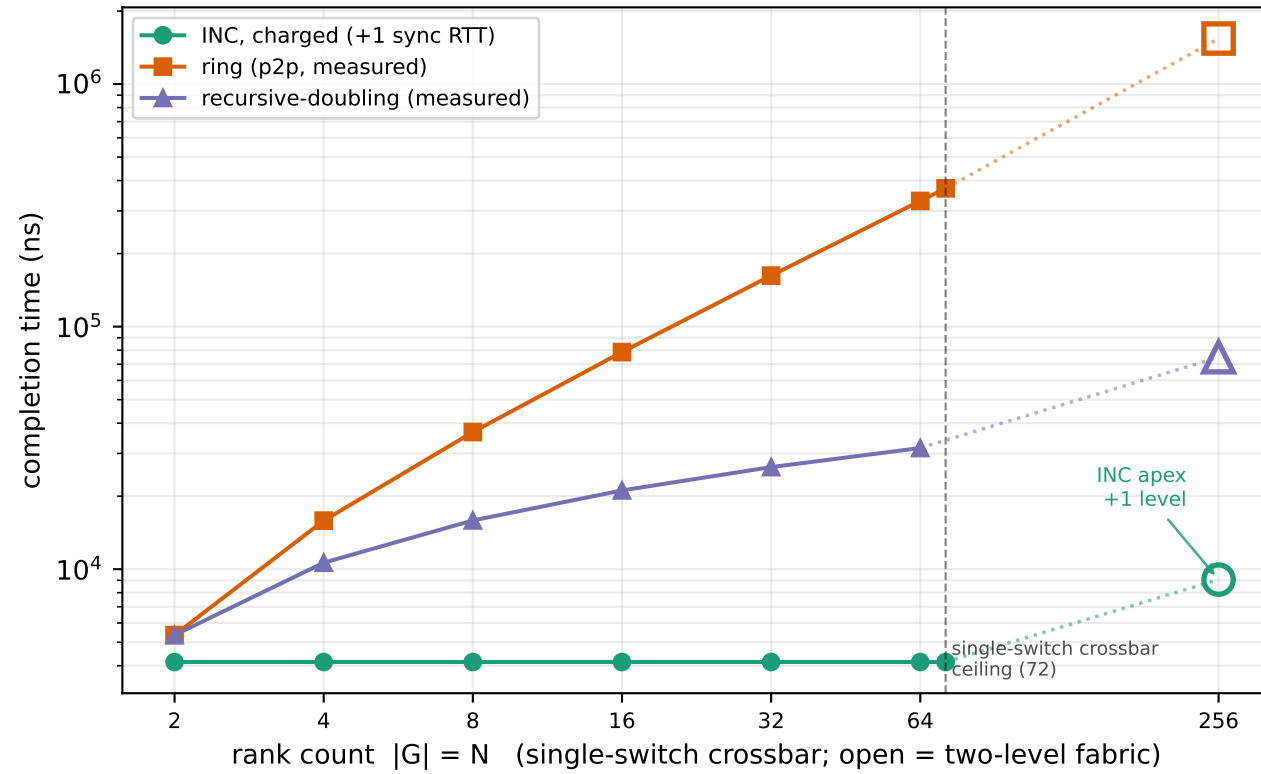
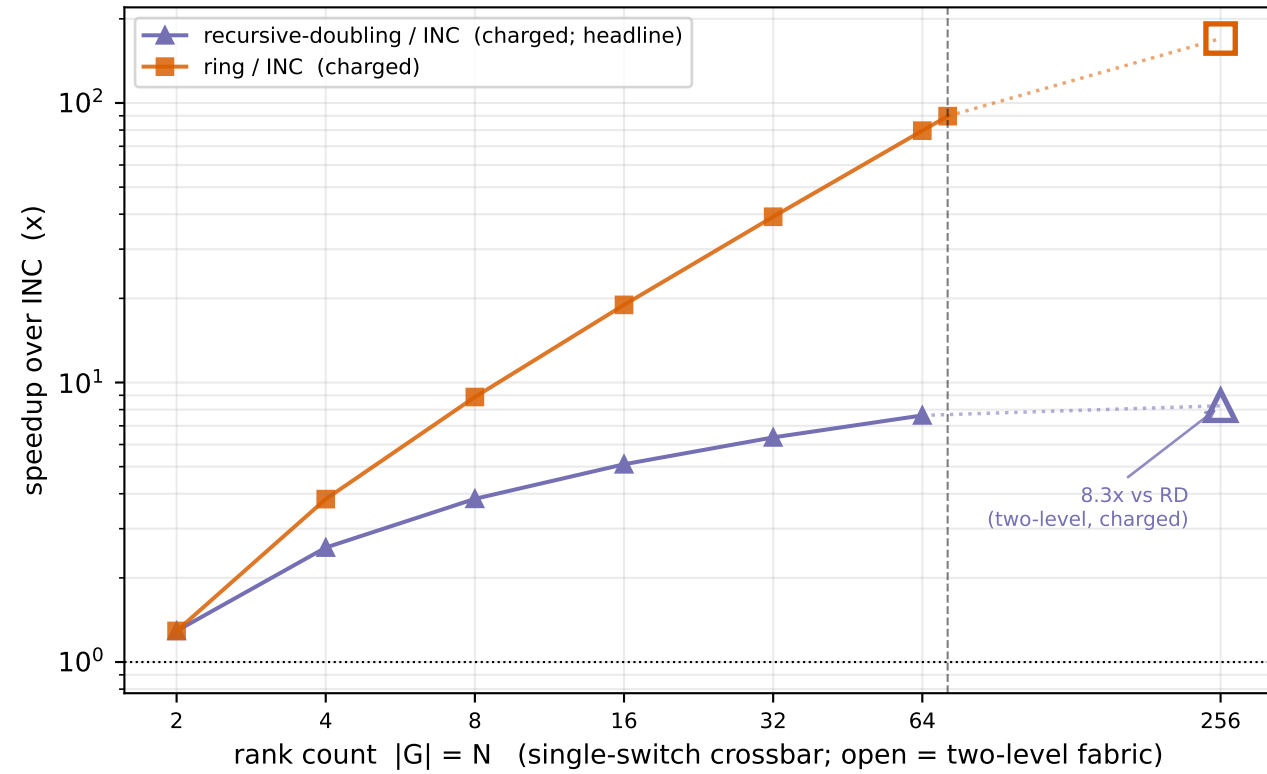


In-network vs point-to-point AllReduce scaling in |G| (htsim_uec fork)
 single-switch scale-up crossbar (radix=|G|, 2..72) + 256-GPU two-level fabric (open markers); 3600 Gbps/port, 500 ns link + 300 ns switch;
 all series measured; INC charged +1 sync RTT (2,600 ns crossbar / 5,800 ns two-level)

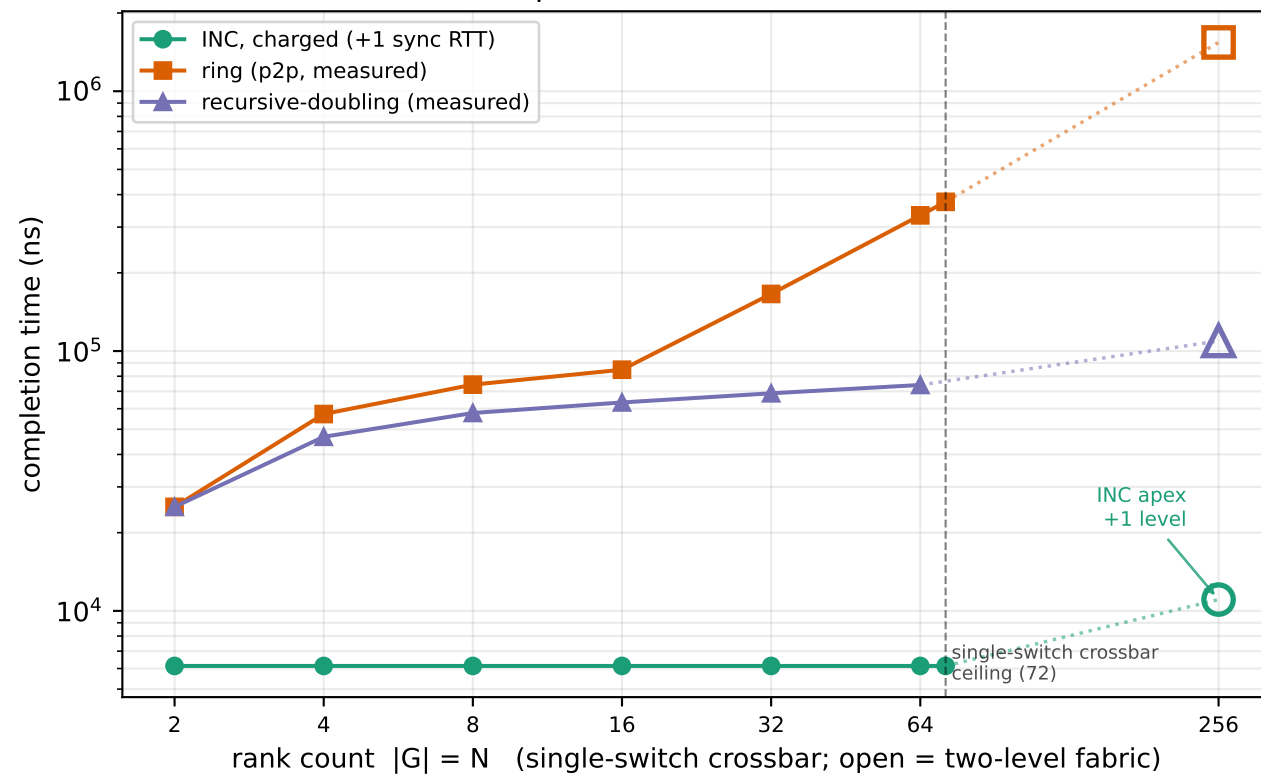
Completion vs N (63KiB/rank)



INC speedup vs N (63KiB/rank)



Completion vs N (1MiB/rank)



INC speedup vs N (1MiB/rank)

